Chapter 3

Ergonomics and Musculoskeletal Issues in Sonography

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What Is Ergonomics?

- Broadly defined, ergonomics is the study of the human body at work.
- Its primary goal is to increase productivity while decreasing worker injury.
- This goal is accomplished by modifying products, tasks, and work environments.

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History of Ergonomics

- Association between work activities and musculoskeletal injuries has been documented for centuries.
- The term *ergonomics* first entered the modern lexicon in 1857.
- In the early 1900s, industry production was still largely dependent on human power and motion, rather than on machines. Ergonomic concepts gained popularity during the Industrial Revolution.

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History of Ergonomics (Cont.)

- Scientific management, which is defined as a method that improved worker efficiency by improving the job process, became popular.
- Assembly line production allowed workers to produce more efficiently by bringing the work to the worker.
- This method improved productivity in the Ford Motor Company but reduced the need of the workers to move throughout their workday, which resulted in static work postures.

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History of Ergonomics (Cont.)

- World War II brought an even greater interest in human-machine interaction.
- Focus of equipment design was expanded to include worker safety, as well as productivity.

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Work-Related Musculoskeletal Disorders (WRMSD) in Sonography

- Awareness of pain and discomfort associated with the occupation of sonography surfaced around 1980 just before the widespread use of real-time scanners.
- Most common complaint was shoulder pain in the sonographer's scanning arm.
- Incidence of WRMSDs was found to be 81% in the United States and 87% in Canada for a combined average incidence in North America of 84%.

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Occupational Safety and Health Act

- 1970: Congress passes the federal Occupational Safety and Health Act.
- Purpose of this Act is to ensure, as far as possible, that every working man and woman in the nation has safe and healthful working conditions.
- Act is administered by the Occupational Safety and Health Administration (OSHA), an agency of the US Department of Labor.

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Occupational Safety and Health Act (Cont.)

- Approximately one half of the states opted to be regulated by OSHA.
- Remaining states created their own agencies that operate under individual state plans.

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General Duty Clause

- Where industry-specific guidelines do not exist, the general duty clause is used.
- Criteria for applying the general duty clause
  - No acceptable standard exists for an industry.
  - Exposure to a hazard that causes serious physical harm exists.
  - Hazard is recognized by the industry.
  - Feasible abatement method exists to correct the hazard.

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General Duty Clause (Cont.)

- Section 5B to the general duty clause states that each employee shall comply with occupational safety and health standards and all rules, regulations, and orders issued pursuant to this Act that are applicable to his or her own actions and conduct.

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Terms for Occupational Injury

- Cumulative trauma disorder (CTD)
- Repetitive motion injury (RMI)
- Overuse syndrome
- Repetitive strain injury (RSI)
- Musculoskeletal strain injury (MSI)
- Work-related musculoskeletal disorder (WRMSD)
  - This is the term that is currently in use.

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WRMSD Incidents

- WRMSD incidents are defined as injuries that:
  - Result in restricted work.
  - Result in days away from work.
  - Involve MSD symptoms that remain for 7 or more days.
  - Involve MSD symptoms that require medical treatment beyond first aid.

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WRMSD and Sonography

- Ultrasound examination specialties such as echocardiography, high-risk obstetric scanning, and, to a lesser extent, vascular sonography involve repetitive motion that can cause WRMSD.

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Alliance Program

- The Alliance Program helps organizations work with OSHA to prevent workplace injuries by educating and leading employers and their employees in advancing workplace safety and health.

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International Ultrasound Industry Consensus Conference

- Organized in May 2003
- Society for Diagnostic Medical Sonography (SDMS) hosts International Ultrasound Industry Consensus Conference
- Goal: To develop injury risk-reducing standards to address the problem of WRMSDs in sonography

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International Ultrasound Industry Consensus Conference (Cont.)

- Administrators addressed the issues of workload, scheduling, and room size.
- Sonographers discussed best practices, education, and training.
- Also discussed were the need for accredited programs to include curriculum related to ergonomics and injury prevention and the need for certifying bodies to test knowledge of risk factors.

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International Ultrasound Industry Consensus Conference (Cont.)

- The industry standards address the role of employees and employers, educators, medical facilities, and equipment manufacturers in reducing the impact of work-related injuries on the workforce and are intended to assist all stakeholders in making informed decisions.

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Injury Data in Sonography

- WRMSDs
  - Include injuries of muscles, tendons, and joints
  - Is defined as injuries that are caused or aggravated by workplace activities
  - Is the main reason for long-term absences among health care workers
  - Account for up to 60% of all workplace illnesses
  - More than 80% of sonographers have some form of MSDs that can be attributed to their work activities

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Injury Data in Sonography (Cont.)

- Factors that contribute to reported injury rates
  - Worker awareness
  - Unwillingness to work in pain
  - Busy patient schedules
  - Job dissatisfaction
  - Aging workforce
  - Computerization of the workplace

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Injury Data in Sonography (Cont.)

- Positive relationship is demonstrated between the severity of MSDs and the performance of repetitive work task(s) that require forceful movements, with or without repetitive motion.
- Increasing use of technology resulted in workers able to accomplish the same work tasks with fewer movements.
- Relationship between the user and the work station equipment has become “frozen,” and the worker is often forced into a static posture.
- Combination of repetitive motions and prolonged static postures results in musculoskeletal discomfort and, eventually, injury.

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Risk Factors for WRMSDs

- Risk factors
  - Forceful exertions
  - Awkward postures and prolonged static postures
  - Repetitive motions
  - “Pinch” grip
  - Environmental factors such as extreme heat, cold, humidity, or vibrations

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Mechanisms of Injury

- Sustained awkward postures can cause imbalances between the muscles that move and the muscles that stabilize.
- Repeatedly rotating the head, neck, and trunk causes one set of muscles to become stronger and shorter and the opposing muscles to become weaker and elongated.

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Mechanisms of Injury (Cont.)

- Asymmetric forces are exerted on the spine, causing misalignment.
- Nerve entrapment syndromes can result from increased muscle and tendon pressure on major nerves that run behind tightened muscles.
- Tasks that require the worker to lean forward continually or to bend the head down or laterally are examples of these types of postures.

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Mechanisms of Injury (Cont.)

- Prolonged static postures, whether sitting or standing, increase the load on soft tissues and the compressive forces on the spine.
- Contraction of more than 50% of the body's muscles is required to maintain static postures.
- Human physiologic functioning depends on movements that promote normal muscle contraction and relaxation.

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Mechanisms of Injury (Cont.)

- Muscle activity circulates blood to carry nutrients to and remove toxins from the muscles.
- Awkward and static postures cause muscles to be contracted continuously; therefore they cannot receive oxygen or get rid of toxins.

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Types of Injury

- Tendonitis and tenosynovitis: Inflammation of the tendon and the sheath around the tendon, which often occur together.
- de Quervain disease: Specific type of tendonitis of the thumb that can result from gripping the transducer.

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Types of Injury (Cont.)

- Carpal tunnel: Entrapment of the median nerve as it runs through the carpal bones of the wrist. Results from repeated flexion and extension of the wrist and also develops from mechanical pressure against the wrist.
- Cubital tunnel: Entrapment of the ulnar nerve as it runs through the elbow. Can result from repeated twisting of the forearm and mechanical pressure against the elbow as it rests on the examination table during scanning.

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Types of Injury (Cont.)

- Epicondylitis (lateral and medial): Inflammation of the periosteum in the area of the insertion of the biceps tendon into the distal humerus. Can result from repeated twisting of the forearm
- Thoracic outlet syndrome: Nerve entrapment that can occur at different levels, resulting in a variety of symptoms
- Trigger finger: Inflammation and swelling of the tendon sheath in a finger that entraps the tendon and restricts the motion of the finger

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Types of Injury (Cont.)

- Bursitis (shoulder): Inflammation of the shoulder bursa from repeated motion.
- Rotator cuff injury: Repeated motion, which causes fraying of the rotator cuff muscle tendons. This injury increases with age and is even more prevalent when work-related stresses are added. Repeated arm abduction contributes to this injury by restricting blood flow to the soft tissues of the shoulder.

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Types of Injury (Cont.)

- Spinal degeneration: Intervertebral disk degeneration that results from bending and twisting and improper seating.

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Industry Awareness and Changes

- Productivity increases if an employee's work area is arranged for the individual worker and the type of work being performed.
- Solutions to occupational injury among sonographers require a combined effort on the part of equipment companies, employers, and sonographers.

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Industry Awareness and Changes (Cont.)

- Engineering solutions
  - Involve a change in the physical features of a workplace
  - Are the preferred methods for control because they can effectively eliminate workplace hazards
  - Tend to be the most expensive initially

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Industry Awareness and Changes (Cont.)

- Administrative solutions
  - Involve workplace policies, changes in patient scheduling and sonographer rotations, and the implementation of rest breaks
  - Control and lessen the duration and frequency of exposure to an injury risk

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Industry Awareness and Changes (Cont.)

- Personal protective equipment (PPE)
  - Use of PPE and/or professional practices is promoted.
  - Addresses best practices and the use of arm support devices
  - Although sonographers are still exposed to the risk factor, the exposure is somewhat reduced.
  - Least effective control

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Ultrasound Systems

- System should be easily mobile and have brakes.
- Control panel should be height adjustable and swivel.
- Monitor should also be height adjustable, independent of the control panel and able to turn and tilt.

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Ultrasound Systems (Cont.)

- Controls should be easy to access without over reaching.
- Transducers should be neither too wide (which causes stretching of the fingers) nor too narrow (which causes a “pinch grip”).
- Transducer cables should be thin, flexible, and lightweight; the transducer cable should be supported during an examination.
- Transducers should be easy to activate with readily accessible connecting ports and storage.

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Other Equipment

- Examination table should be electronically height adjustable.
- Examination table should also be specialty-specific, with options that adapt it for use in certain procedures.
- If the sonographer sits to scan, a height-adjustable chair with an appropriate height range is equally as important as the examination table.

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Other Considerations

- Sonographers should be able to support their arms while scanning.
- Examination room should be large enough to allow for a flexible set up.
- Examination room must have appropriate lighting to reduce the glare on the monitor and subsequent eye strain.

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Administrative Controls

- Schedule patient examinations to prevent the repetition of the same type of examination back to back.
- Have sonographers perform a variety of examinations, allowing different muscles to fire.
- Design the schedule to allow enough time between examinations for muscle recovery.
- Examination gloves should have textured fingers to prevent the need to grip the transducer too tightly.
- Recommend that sonographers take short “mini” breaks during examinations to relax muscles, especially in the shoulder and neck.

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Administrative Controls (Cont.)

- Bedside examinations
  - Ensure that the ultrasound system can be moved easily and has a small footprint.
  - Try to share bedside examinations with other staff
  - Bedside examinations should be reserved for those patients whose condition prohibits transporting them.

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Administrative Controls (Cont.)

- Provide separate monitors so that the patient and the sonographer do not have to share the monitor mounted on the system.
- Provide ergonomically designed scanning rooms to reduce the risk of injury to the sonographer, including appropriately adjustable ancillary equipment.

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PPE and Professional Controls

- Minimize sustained bending, twisting, reaching, lifting, and transducer pressure.
- Avoid awkward postures.
- Alternate sitting and standing throughout examinations.
- Vary scanning techniques and transducer grips.
- Adjust all equipment to suit each user’s size.
- Have accessories on hand before beginning the examination.
- Use appropriate measures to reduce arm abduction.

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PPE and Professional Controls (Cont.)

- Avoid forward and backward reach.
- Instruct the patient to move as close to you as possible.
- Adjust the height of the table and chair.
- Use support for your arms.
- Relax your muscles periodically throughout the day.
- Stretch your hand, wrist, shoulder muscles, and spine.
- Take mini breaks during the procedure.

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PPE and Professional Controls (Cont.)

- Take meal breaks separate from work-related tasks.
- Using the 20-20-20 rule, refocus your eyes every 20 minutes on an object approximately 20 feet from you for approximately 20 seconds.
- Vary procedures, tasks, and skills as much as reasonably possible.
- Use correct body mechanics when moving patients, wheelchairs, beds, stretchers, and ultrasound systems.
- Report and document any persistent pain to your employer, and seek competent medical advice.

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PPE and Professional Controls (Cont.)

- Maintain a good level of physical fitness to perform the demanding work tasks required.
- Collaborate with employers on staffing solutions that allow sufficient time away from work.

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Work Practice Changes

- Gripping the transducer
- Flexing and extending the wrist
- Twisting the neck
- Abducting the scanning arm
- Managing the transducer cable
- Twisting the trunk
- Reaching

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Economics of Ergonomics

- Cost of occupational injury to both the employer and the employee is phenomenal.
- Losses to the employer encompass not only the medical costs of an injury, but also the cost of replacement staff, worker's compensation, and loss of revenue.

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Economics of Ergonomics (Cont.)

- Loss of experienced professionals and a skilled, stable workforce also affect productivity.
- Cost to the worker includes not only monetary hardship, but also the possibility of permanent injury, chronic pain, and loss of profession.

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Economics of Ergonomics (Cont.)

- MSDs are also the most costly of all occupational problems, accounting for the majority of worker's compensation costs.
- $1 of every $3 of worker's compensation costs are spent on occupational MSDs.
- Employers pay $15 to $20 billion per year in worker's compensation costs for lost workdays.
- Mean cost per cause of upper extremity WRMSD is $8070 versus a mean cost of $4075 per case for all types of work-related injury.

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Economics of Ergonomics (Cont.)

- Indirect costs are three to five times higher, reaching approximately $150 billion per year.
  - Absenteeism, staff replacement and retraining, and loss of productivity and/or quality
- Cost of hiring temporary replacement staff is between $130,000 and $166,000 per year.
- Estimated average cost to find and hire a new sonographer is $10,000.

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Economics of Ergonomics (Cont.)

- Cost of equipping a sonography examination room is minimal compared with the cost of a worker's compensation injury.
- Quality of the patient's examination may also suffer if the sonographer is in pain while performing the examination.

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Economics of Ergonomics (Cont.)

- Accessory equipment that mitigates injury risk include:
  - Height adjustable stool
    - Cost is $750 reimbursement in two to three patient studies.
  - Set of support cushions
    - Cost is $250 reimbursement in one patient study.
  - Ergonomic examination table
    - Cost is $7400 reimbursement from 2 days' work

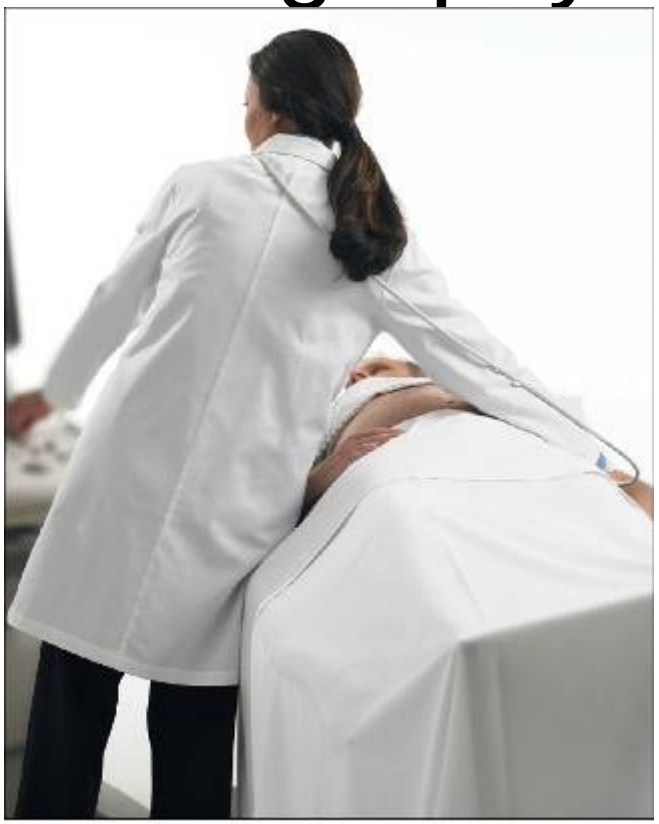
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Economics of Ergonomics (Cont.)

- Work-related injuries must be reported immediately to occupational health or risk management departments.
- Injuries should be recorded on OSHA logs.
- Failure to do so may result in the denial of claims.

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Ergonomic Practices in Sonography



(Courtesy Philips Healthcare, Ultrasound, North America.)

Asymmetric forces are exerted on the spine when more weight is placed on one side of the body than the other. The head must be turned to view the monitor, resulting in a twisted neck.

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Ergonomic Practices in Sonography (Cont.)



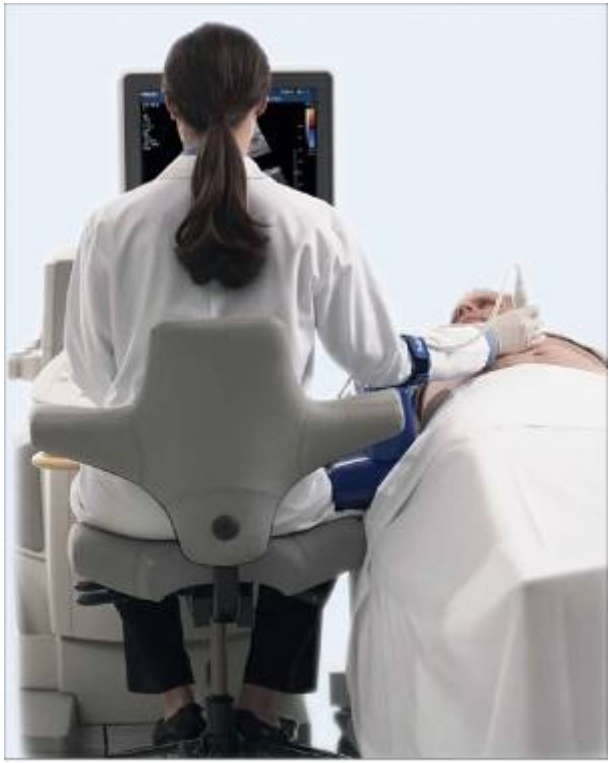
A

(Courtesy Susan Rantz Stephenson.)

You must take responsibility for poor postural alignment.

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Ergonomic Practices in Sonography (Cont.)



B

(Courtesy Philips Healthcare, Ultrasound, North America.)

Sonographer sits up straight with the top of the monitor level with the eyes and an arm's length away. The feet are supported on the ring of the chair, and the forearm is supported on a cushion.

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Ergonomic Practices in Sonography (Cont.)



A

(Courtesy Siemens Healthcare, Ultrasound USA Division.)

Bad ergonomics. Right-handed cardiac scanning results in the abduction of the arm over the patient's back, hyperflexion of the right wrist, asymmetric posture, and twisting of the neck.

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Ergonomic Practices in Sonography (Cont.)



B

(Courtesy Siemens Healthcare, Ultrasound USA Division.)

Good ergonomics. Turning the patient around to perform the study is one way to reduce risk factors.

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Ergonomic Practices in Sonography (Cont.)



A

(Courtesy Sound Ergonomics, LLC, Kenmore, Washington.)

Bad ergonomic practice in left-handed cardiac scanning.

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Ergonomic Practices in Sonography (Cont.)



B

(Courtesy Sound Ergonomics, LLC, Kenmore, Washington.)

Good ergonomics in left-handed cardiac scanning. The height of the chair is adjusted up and the table down to reduce the angle of abduction to 30 degrees or less. The patient is then brought close to edge of the table.

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Ergonomic Practices in Sonography (Cont.)

Bad vascular ergonomics



A

(Courtesy Sound Ergonomics, LLC, Kenmore, Washington.)

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Ergonomic Practices in Sonography (Cont.)



B

(Courtesy Siemens Healthcare Ultrasound, USA Division.)

When scanning the patient's left leg, the patient is turned so that the patient's left leg is closest to the sonographer.

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Ergonomic Practices in Sonography (Cont.)



C

(Courtesy Sound Ergonomics, LLC, Kenmore, Washington.)

The most ergonomic way to perform a scan for the evaluation of DVT is shown if the patient is mobile.

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Ergonomic Practices in Sonography (Cont.)

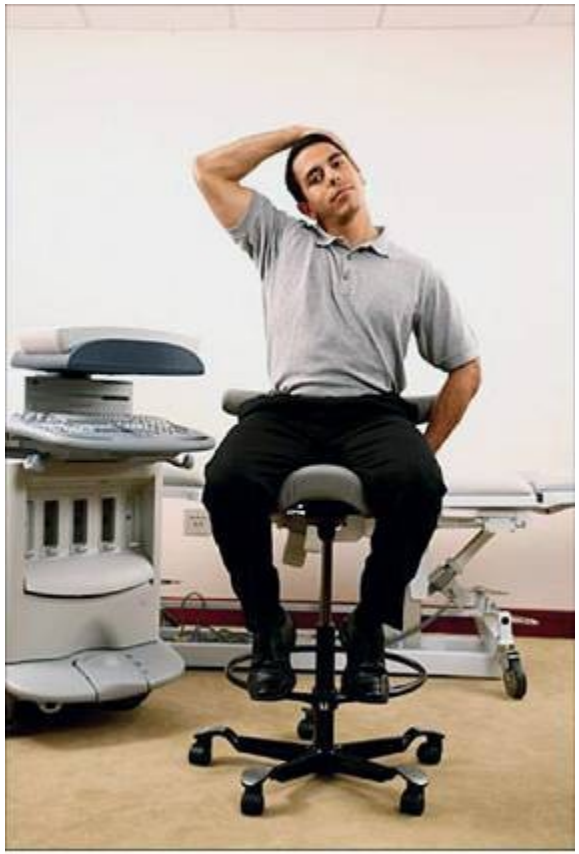


(Courtesy Sound Ergonomics, LLC, Kenmore, Washington.)

For transvaginal scanning, twisting of the forearm can be reduced or avoided by scanning from the foot of the table with the patient in stirrups.

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Ergonomic Practices in Sonography (Cont.)



A

(Courtesy Siemens Healthcare Ultrasound, USA Division.)

Simple stretches and exercises that are performed throughout the day for a few minutes make a significant difference to health and well-being.

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Ergonomic Practices in Sonography (Cont.)



B

(Courtesy Siemens Healthcare Ultrasound, USA Division.)

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